Md Shahanur Islam Shagor

Full-Stack Developer & AI Engineer

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I am writing to apply for the Ground Control Station Software Engineer - N3XT Interceptor C-UAS position at Quantum-Systems GmbH. Your work on complete operational systems that integrate aircraft, ground control stations, sensor inputs, and communication links is exactly the kind of real-world engineering challenge I want to be part of.

Through research and development work, I have built systems that combine software engineering with hardware interaction, real-time data processing, and autonomous operation. My work on the Wireless Vision-Aid for the Blind (WVAB) involved developing a real-time object detection pipeline using YOLOv8n on a Raspberry Pi with ESP32-CAM, implementing ultra-low latency UDP streaming for video transmission, and integrating multilingual audio feedback. This project required designing robust software that operates reliably under field conditions, processing live sensor data with minimal latency.

One area that particularly interests me is building frontend and backend systems that interact with live data streams and hardware. I have experience developing real-time communication systems using WebSocket and Socket.IO, and I have worked with Linux-based environments, Git, CI/CD, and Docker for deployment. My professional experience includes building RESTful APIs, implementing authentication and role-based access control, and optimizing database performance. I am comfortable working with C++, TypeScript, Python, and Go, and I understand the importance of low-latency data routing, telemetry handling, and robust error management.

In addition to my technical experience, I work effectively across multidisciplinary teams and communicate clearly with both technical and non-technical stakeholders. I am fluent in Bengali and English, have working knowledge of Russian and Hindi, and enjoy collaborating in international and cross-cultural environments.

I have attached my CV and cover letter for your review. I would welcome the opportunity to discuss how my background in building real-time, hardware-interfacing systems can contribute to the N3XT Interceptor program.

Sincerely,

Md Shahanur Islam Shagor
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